

A car rental company operates a number of rental locations throughout the metropolitan area. Since the company has a great business model and provides customer-friendly service, its business has boomed over the last several years. As the business has grown rapidly, the costs of running its business has also increased. In particular, as the job market becomes hot, the labor cost has doubled over the last several years. The company wants to find a solution to reduce its operating cost. The business operation of the company is described as follows. The description is not meant to be complete, and the company is flexible enough to consider any good improvement proposals.

Vehicles can be taken from one location and returned to the same location or to a different location with an additional charge. Although the company is, at present, concerned only with passenger cars, it may branch out into other forms of vehicle rentals in the future and would like to be able to use the same reservation system. The company has several different makes of cars in its rental fleet, from different manufacturers. Each make may have several models. For example, Toyota has Corolla, Camry, and more. The models are grouped into a small number of price classes. The customer must be able to select the make and the model he or she wants to rent. If the selected car is not available, the system should display a message telling the customer that the car is rented out and let the customer select another make and model, or have the system suggest similar models of a different make.

The company has a number of different rental plans available to customers. For example, there is a “daily unlimited miles plan” and a “weekend 10% discount plan.” The company finds it important to have information available on the models of car, such as automatic or manual gear change, two or four doors, and sedan or hatchback. The rental prices may be different for different options and a customer will want to know such information when reserving a car. Currently, customers make reservations directly with the car rental company either in person or through the phone. The salespersons process the reservations manually using a reservation form and archive them in the file cabinet. No deposit is required at the time of reservation. The reservation is voided if the customer does not show up to sign the contract for more than a given period of time. Such reservation is honored only if there are still cars available to satisfy the request.

Sometimes a customer wishes to make a block reservation for several cars and to have the invoices for all rentals on the reservation handled together. As soon as a

car is checked out to a customer, an invoice is opened. A single invoice may cover one or more rentals. Normally a customer will settle the invoice when the car is returned but, in some cases, the invoice must be sent to a company (such as the customer's employer). When the customer pays by a credit card, the rental charge will be processed through a credit card processing company.

A car may or may not be available for rental on a given day. Rental cars need frequent preventive maintenance and, in addition, any damage to a car has to be repaired as soon as possible. The company wants to keep track of the rental car purchase, repair, maintenance, and disposal information for business and tax purposes (e.g., depreciation of the rental cars).